

# Final Report: Mach-Zehnder Interferometer

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**Abstract**— In this work we provide a design of Mach-Zehnder interferometer and analyse the effect of interferometer characteristics. The parameters analysed include waveguide geometry, waveguide mode profiles, transfer function of the interferometer, Corner Analysis, and others. Initially, the design and simulations are implemented in Lumerical MODE and Lumerical INTERCONNECT programs. Then, the fabricated design is tested and experimental measurements were analysed.

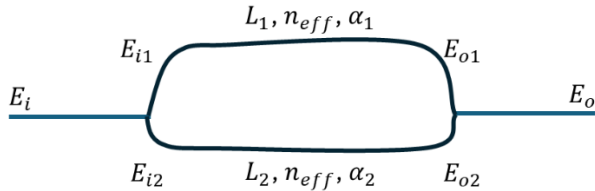
**Index Terms**— *interferometers, silicon photonics, simulations.*

## I. INTRODUCTION

THE recent decades were prominent for development of telecom and datacom and related use of photonic components for such applications. Some of the advantages of silicon photonics are scalability and compatibility with CMOS process lines [1]. In this context, high-speed switches with low crosstalk are in a high need [2]. For broadband switch design, one of the widely used types is Mach-Zehnder interferometer (MZI). Considering the high sensitivity of silicon photonics interferometric devices to few-nanometer variations in geometry [3], it is important to carefully simulate and design the MZI to obtain the desired device characteristics. In this regard, in this work we provide theoretical foundations of MZI in Section II, then we provide waveguide modelling methodology and results in Section III, after Section IV covers Fabrication aspects, Experimental data and Analysis will be provided in Sections V and VI correspondingly. At the end, conclusion of the work will be provided.

## II. THEORY OF MACH-ZEHNDER INTERFEROMETRY

Mach-Zehnder interferometer measures the relative phase shift between two optical paths. The main principle lies in splitting the input beam into two beams using beam splitter, two beam paths can have the same lengths (balanced MZI) or different one (unbalanced MZI). After the beams are combined using beam combiner and the resulted output can be analysed.



**Fig. 1.** Schematics of Mach-Zehnder interferometer.

The schematics of MZI is shown in Fig.1. The left part of the MZI consists of input beam with electric field  $E_i$ , which splits to electric fields  $E_{i1}$  and  $E_{i2}$  on corresponding arms of MZI:

$$E_{i1} = E_{i2} = \frac{1}{\sqrt{2}} E_i \quad (1)$$

After, both beams travel along the waveguide and affected by losses and phase shift:

$$E_{o1} = E_{i1} \times e^{-j\beta_1 L_1 - 0.5\alpha_1 L_1} \quad (2)$$

$$E_{o2} = E_{i2} \times e^{-j\beta_2 L_2 - 0.5\alpha_2 L_2} \quad (3)$$

where propagation constant  $\beta$  is equal to:

$$\beta_k = \frac{2\pi}{\lambda} n_{eff,k}$$

where subscript  $k$  corresponds to  $k^{th}$  arm of the MZI.

After, the beams are combined in the output, but field is split evenly between fundamental mode and high-order modes. Therefore, the output field is equal to:

$$E_o = \frac{1}{\sqrt{2}} (E_{o1} + E_{o2}) = \frac{1}{\sqrt{2}} \frac{1}{\sqrt{2}} (E_{i1} \times e^{-j\beta_1 L_1 - 0.5\alpha_1 L_1} + E_{i2} \times e^{-j\beta_2 L_2 - 0.5\alpha_2 L_2}) \quad (4)$$

Since the light intensity is proportional to square of field amplitude,  $I \sim E^2$ , the resulted ratio of output and input intensities for MZI is:

$$\frac{I_o}{I_i} = \frac{1}{4} |e^{-j\beta_1 L_1 - 0.5\alpha_1 L_1} + e^{-j\beta_2 L_2 - 0.5\alpha_2 L_2}|^2 \quad (5)$$

In case of zero propagation losses and equal propagations constants:

$$\frac{I_o}{I_i} = \frac{1}{2} (1 + \cos(\beta(L_2 - L_1))) \quad (6)$$

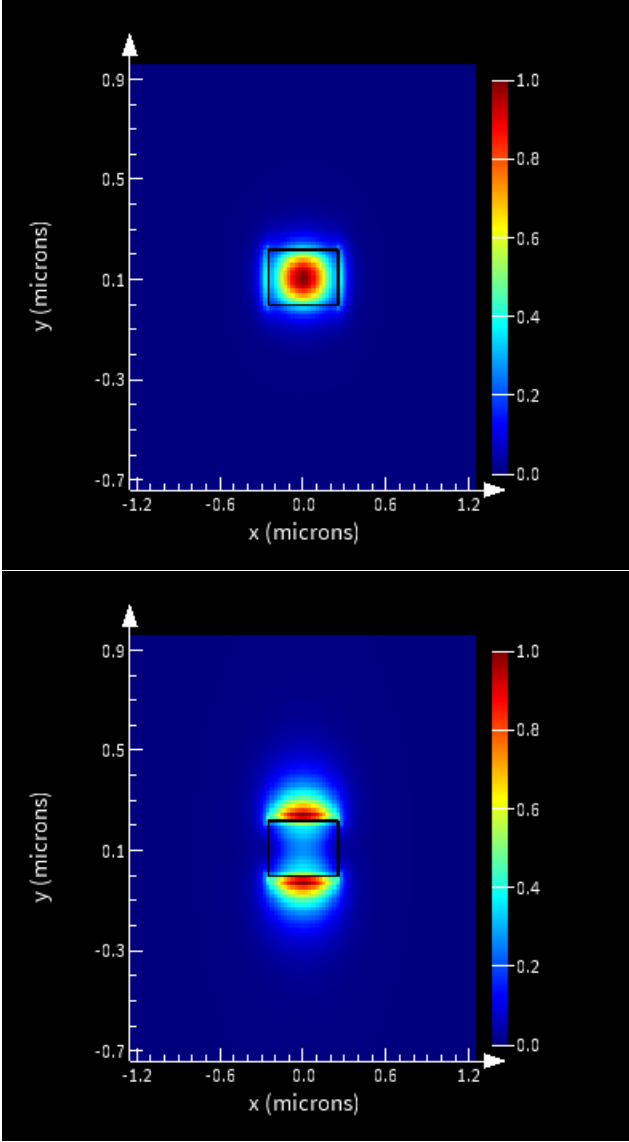
## III. WAVEGUIDE MODELLING

The strip waveguide design has the following dimensions: width of 500 nm, and height of 220 nm. Fig. 2 illustrates the TE and TM modes of the waveguide for 1550 nm wavelength. Figures 3 and 4 report the effective index of the waveguide and group index vs wavelengths correspondingly.

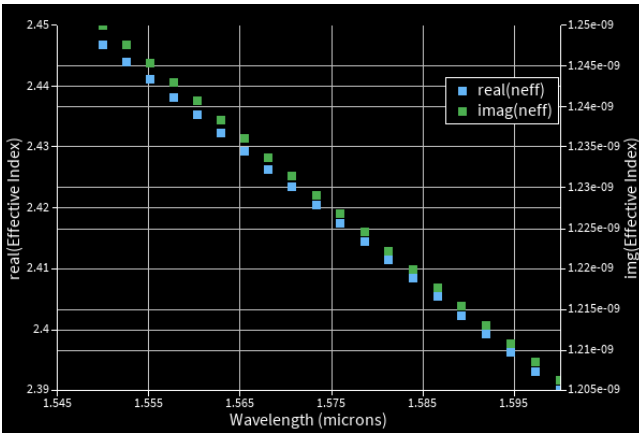
Moreover, the compact waveguide model has been calculated using Lumerical and MATLAB fitting.

$$n_{eff}(\lambda) = 2.446 - 1.133(\lambda - 1.55) - 0.044(\lambda - 1.55)^2 \quad (7)$$

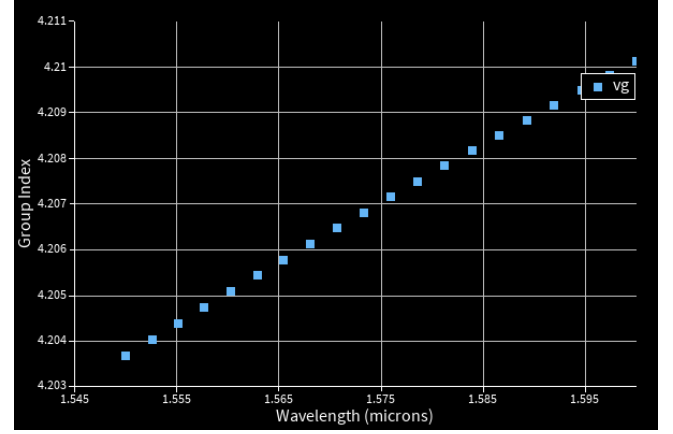
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**Fig. 2.** Electric field intensities in the waveguide for (up) TE mode; (down) TM mode.



**Fig. 3.** Real and imaginary parts of effective index versus wavelength.



**Fig. 4.** Group index vs wavelength.

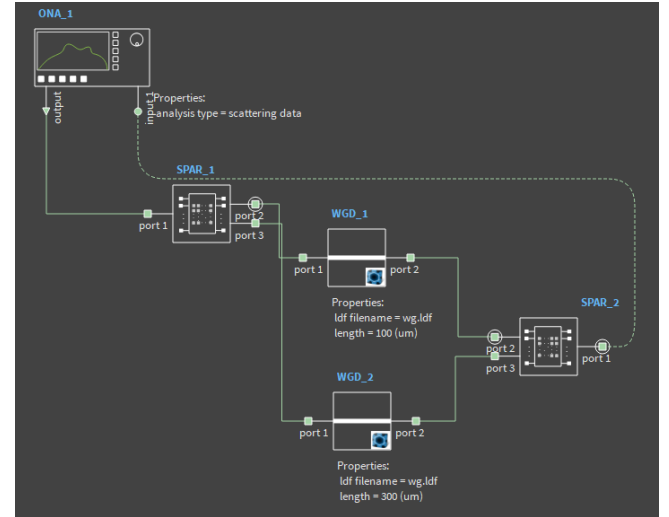
### III. MACH-ZEHNDER INTERFEROMETER

Considering Mach-Zehnder transfer function (Eq. 6), and the Free Spectral Range (FSR) equation:

$$FSR(\lambda) = \frac{\lambda^2}{n_g(\lambda)\Delta L}$$

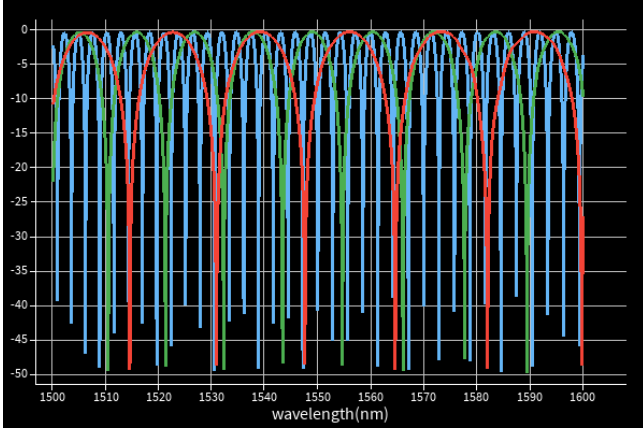
We can analyse the MZI for different  $\Delta L$ :

| L1 [um] | L2 [um] |
|---------|---------|
| 235.953 | 32.938  |
| 83.555  | 32.938  |
| 66.735  | 32.938  |

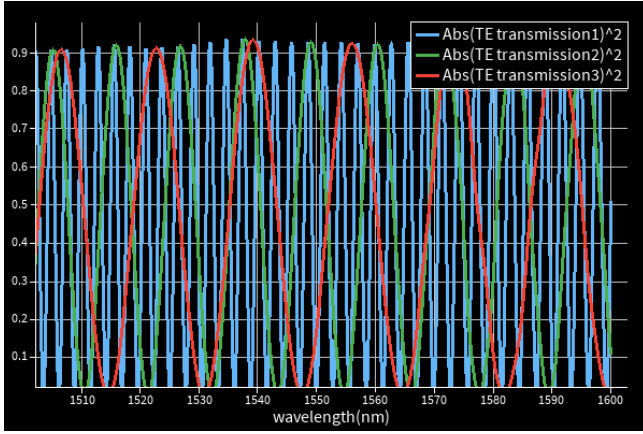


**Fig. 5.** Design of Mach-Zehnder interferometer implemented in Lumerical INTERCONNECT.

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**Fig. 6.** TE gain of MZI for  $L_2=32.938\mu\text{m}$ ,  $L_1=235.953\mu\text{m}$  (blue),  $83.555\mu\text{m}$  (red),  $66.735\mu\text{m}$  (green).



**Fig. 7.** Transmission of MZI for  $L_2=32.938\mu\text{m}$ ,  $L_1=235.953\mu\text{m}$  (blue),  $83.555\mu\text{m}$  (red),  $66.735\mu\text{m}$  (green).

#### IV. CORNER ANALYSIS

For Corner Analysis, to take into account manufacturing variations, different thickness and width values of the waveguide have been simulated: width of 460 nm, 500 nm, and 520 nm; and thickness of 213.3 nm, 220 nm, 225.1 nm were used for Corner Analysis. Wavelength of 1550 nm, and mesh size of 20 nm were used for the simulation parameters.

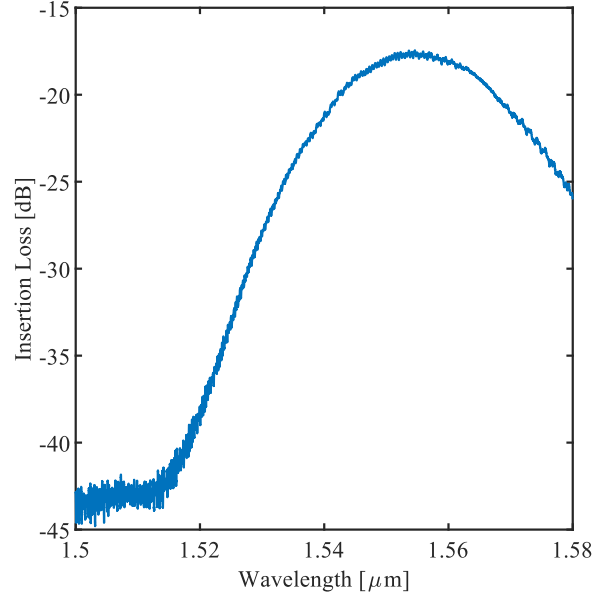
The resulting minimum value of refractive index is 4.13896, and maximum value is 4.26241. The next section presents the experimental results and their relation to the abovementioned Corner Analysis.

#### IV. MEASUREMENTS AND RESULTS

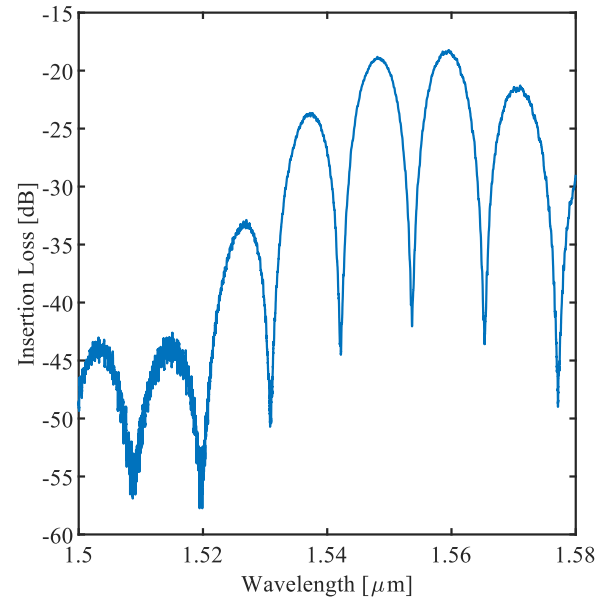
To characterize the devices, a custom-built automated test setup [4, 8] with automated control software written in Python was used [5]. An Agilent 81600B tunable laser was used as the input source and Agilent 81635A optical power sensors as the output detectors. The wavelength was swept from 1500 to 1600 nm in 10 pm steps. A polarization maintaining (PM) fibre was used to maintain the polarization state of the light, to

couple the TE polarization into the grating couplers [6]. A 90° rotation was used to inject light into the TM grating couplers [6]. A polarization maintaining fibre array was used to couple light in/out of the chip [7].

The reference waveguide spectrum is given in Fig. 8. The related effect of grating couplers can be clearly seen. Afterwards, several MZIs were modelled and fabricated. The results of one of them (with path difference of 50.6170  $\mu\text{m}$ ) are given in Fig. 9 - Fig. 12.



**Fig. 8.** Spectrum of the reference waveguide.

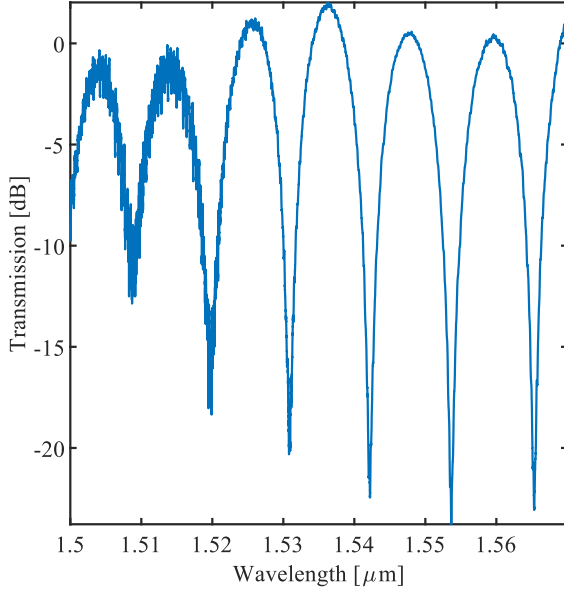


**Fig. 9.** Raw measurements of MZI with path difference of 50.6170  $\mu\text{m}$ .

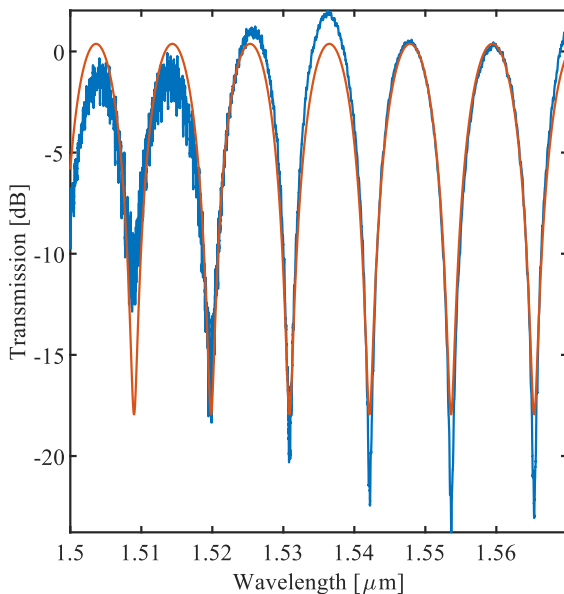
As can be seen, Fig 9 spectrum of MZI is affected by the

grating couplers and waveguide, therefore is it needed to implement baseline correction method utilizing the curve-fitting method. The result of baseline correction is shown in Fig. 10. Then, the fit results are provided in Fig. 11 and allow reconstruction of group index of the system, resulting in Fig. 12.

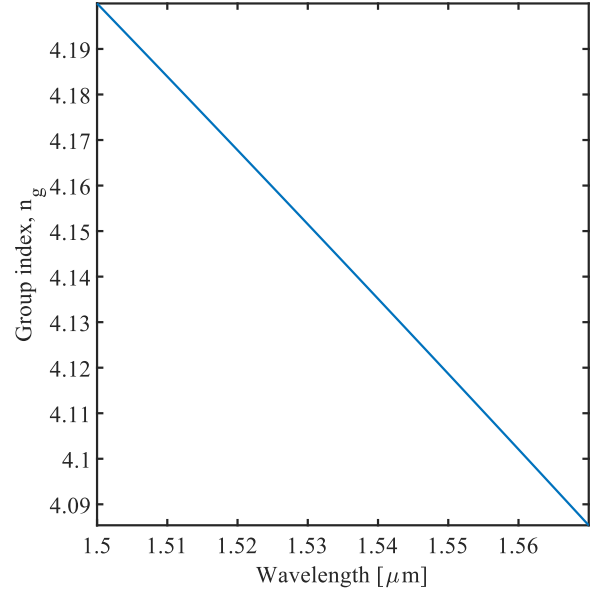
The results show that refractive index obtained is in the range of the Corner Analysis: minimum and maximum group indices are 4.13896 and 4.26241 correspondingly, while the experimental index is 4.1433.



**Fig. 10.** Baseline corrected measurements of MZI with path difference of 50.6170  $\mu\text{m}$ .



**Fig. 11.** Baseline corrected measurements (blue) and MZI model using fit parameters (red).



**Fig. 12.** Group index of MZI fit.

## V. CONCLUSION

In this report, we have provided waveguide modelling and design of Mach-Zehnder interferometer with variation of length difference. The resulted relations between effective index, group index and wavelengths are provided. Moreover, the resulted transmission spectra are given. The simulated results are in accordance with the theoretical expectations. Moreover, Corner Analysis has been performed in order to take in account fabrication variability. The resulted experimental values of the refractive index are in the range of Corner Analysis values.

## VI. SUMMARY OF THE FABRICATION

The photonic devices were fabricated using the NanoSOI MPW fabrication process by Applied Nanotools Inc. (<http://www.appliednt.com/nanosoi>; Edmonton, Canada) which is based on direct-write 100 keV electron beam lithography technology. Silicon-on-insulator wafers of 200 mm diameter, 220 nm device thickness and 2  $\mu\text{m}$  buffer oxide thickness are used as the base material for the fabrication. The wafer was pre-diced into square substrates with dimensions of 25x25 mm, and lines were scribed into the substrate backsides to facilitate easy separation into smaller chips once fabrication was complete. After an initial wafer clean using piranha solution (3:1  $\text{H}_2\text{SO}_4:\text{H}_2\text{O}_2$ ) for 15 minutes and water/IPA rinse, hydrogen silsesquioxane (HSQ) resist was spin-coated onto the substrate and heated to evaporate the solvent. The photonic devices were patterned using a JEOL JBX-8100FS electron beam instrument at The University of British Columbia. The exposure dosage of the design was corrected for proximity effects that result from the backscatter of electrons from exposure of nearby features. Shape writing order was optimized for efficient patterning and minimal beam drift. After the e-beam exposure and subsequent development with a tetramethylammonium sulfate (TMAH) solution, the

devices were inspected optically for residues and/or defects. The chips were then mounted on a 4" handle wafer and underwent an anisotropic ICP-RIE etch process using chlorine after qualification of the etch rate. The resist was removed from the surface of the devices using a 10:1 buffer oxide wet etch, and the devices were inspected using a scanning electron microscope (SEM) to verify patterning and etch quality. A 2.2  $\mu\text{m}$  oxide cladding was deposited using a plasma-enhanced chemical vapour deposition (PECVD) process based on tetraethyl orthosilicate (TEOS) at 300°C. Reflectometry measurements were performed throughout the process to verify the device layer, buffer oxide and cladding thicknesses before delivery.

- [7] [www.plcconnections.com](http://www.plcconnections.com), PLC Connections, Columbus OH, USA.
- [8] <http://mapleleafphotonics.com>, Maple Leaf Photonics, Seattle WA, USA.

#### ACKNOWLEDGEMENTS

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